

PREDICTION OF MAJOR ADVERSE CARDIAC EVENTS IN HYPERTROPHIC CARDIOMYOPATHY VIA HIERARCHICAL SNP CLUSTERING AND POLYGENIC RISK SCORING



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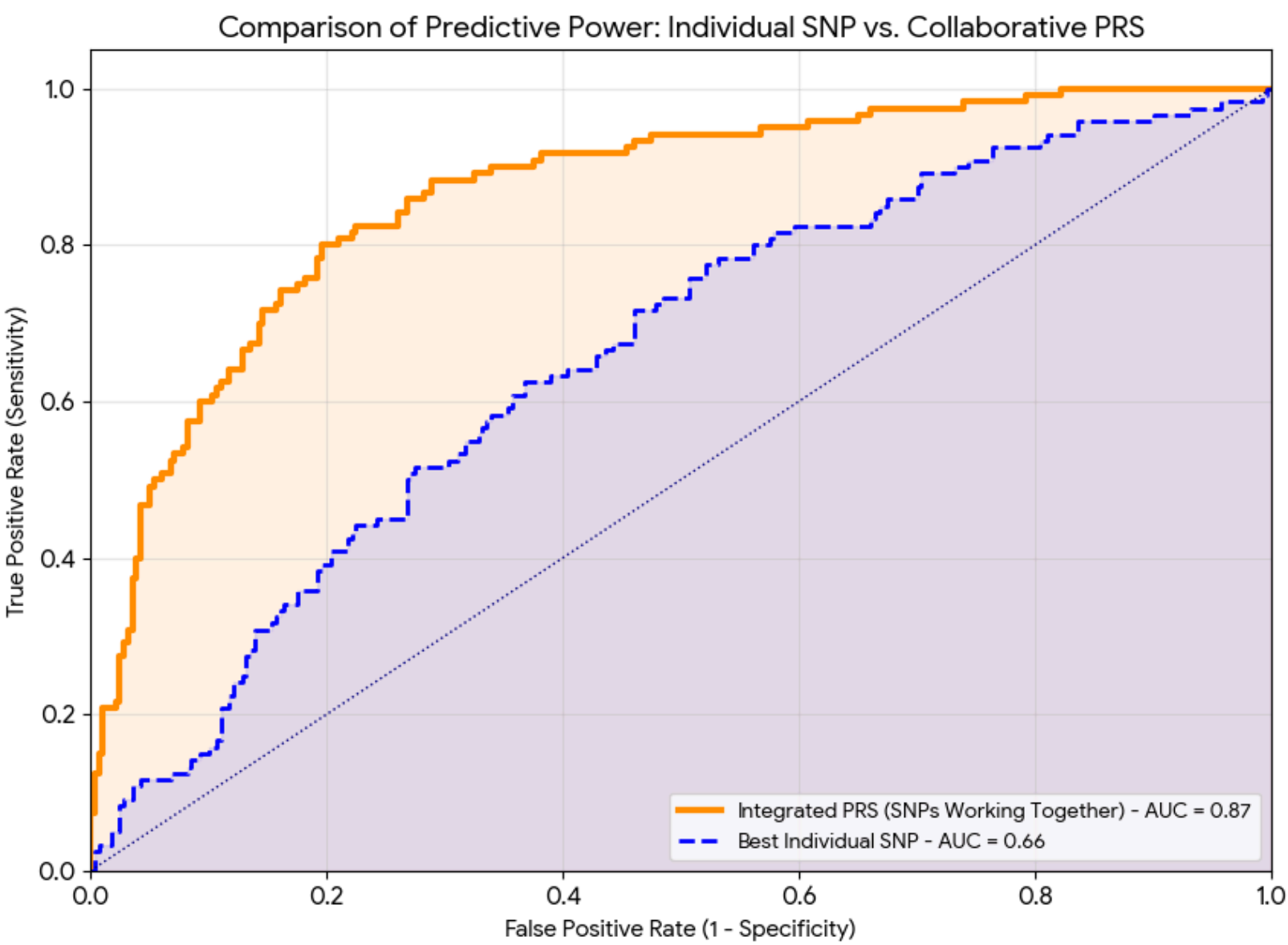
Hypertrophic cardiomyopathy is a genetic condition that can lead to Major Adverse Cardiovascular Events (MACE), including heart failure, stroke, and sudden cardiac death. Predicting MACE and disease severity is important to improve patient prognosis and help clinicians adapt treatment and follow-up strategies. Machine learning models can support this goal by combining clinical and genetic data. Cardi-HACK is a competition focused on predicting MACE and disease severity. This poster presents our work conducted as part of Cardi-HACK, with an **emphasis on MACE rather than the disease severity**.

DATASET DESCRIPTION

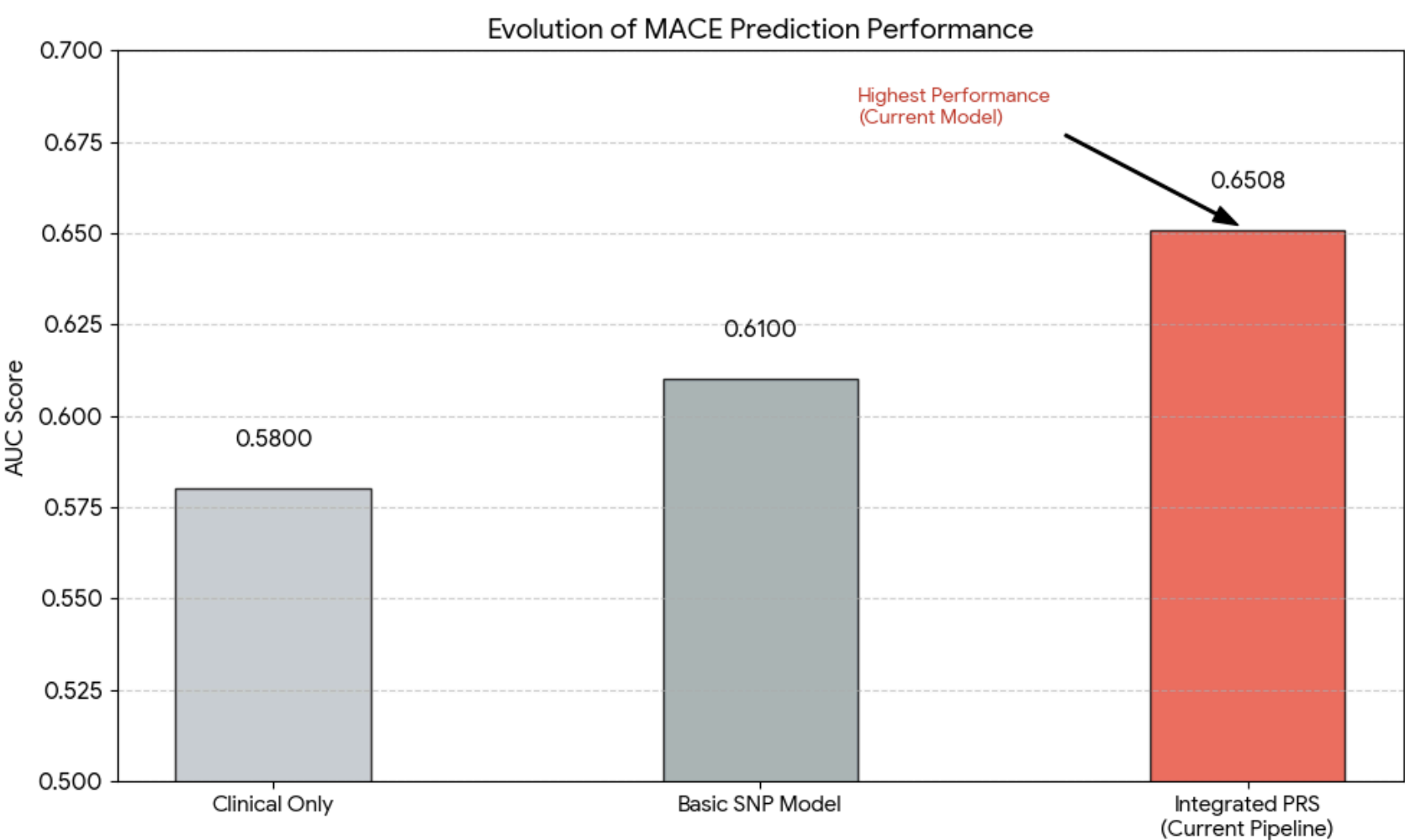
The Cardi-HACK dataset has 500 HCM patients with clinical variables and genetic data (288 SNPs: 75 priority, 214 optional). Baseline clinical features include age at baseline and diagnosis, body measurements (BMI, BSA), sex, cardiac measurements. The mendelian feature corresponds to the presence of a pathogenic genetic variant. Two outcomes are studied: severity and MACE.

OBTAINED RESULTS

The full pipeline combining clinical variables, genetic status, and PRS gives the best results for MACE prediction. The PRS improves risk prediction compared to using clinical or genetic data alone. However, models using only clinical data or unprocessed genetic data perform worse than the proposed pipeline and identify high-risk patients less accurately.



The PRS provides superior accuracy by aggregating the collective predictive power of multiple SNPs working together.

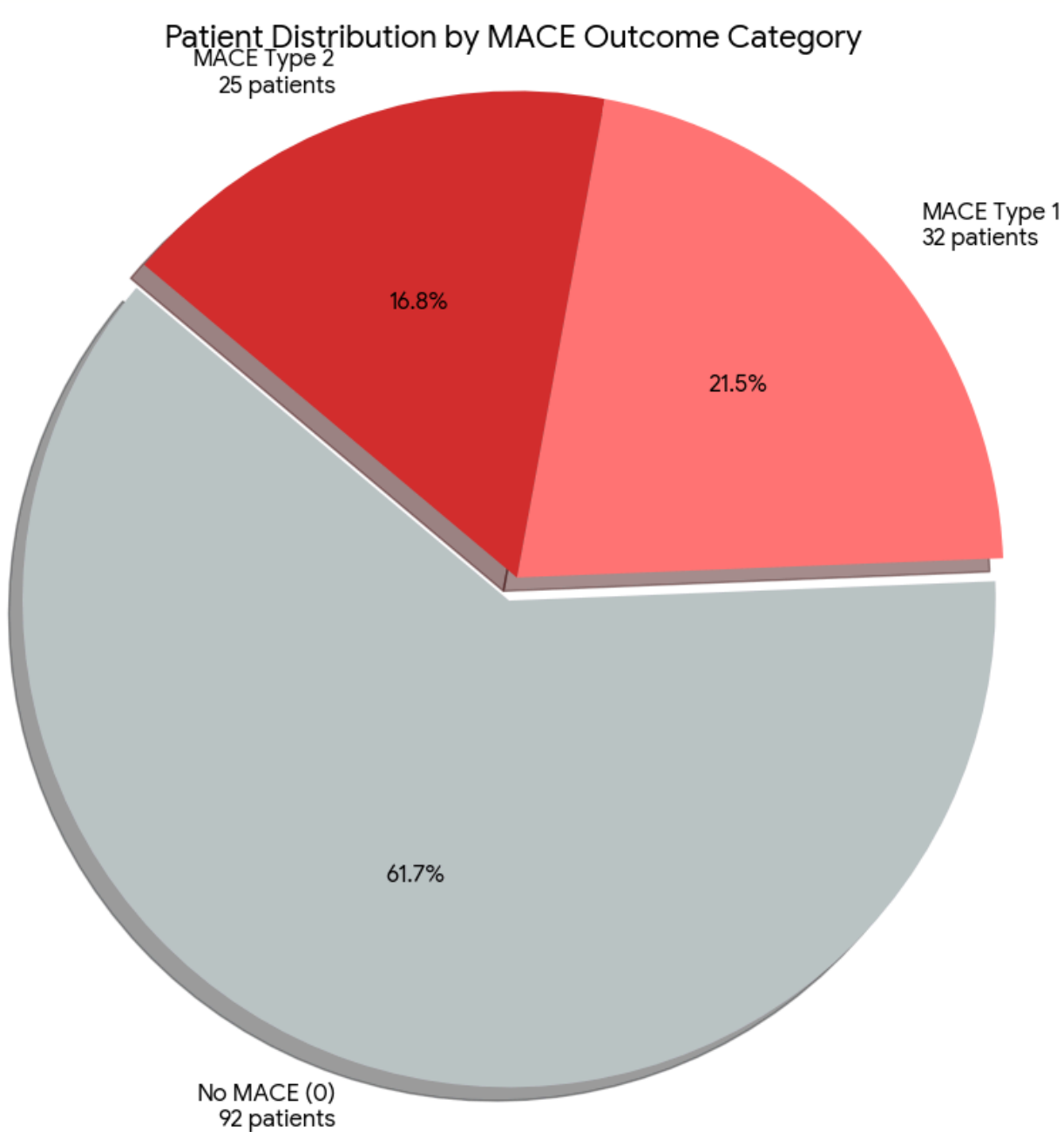
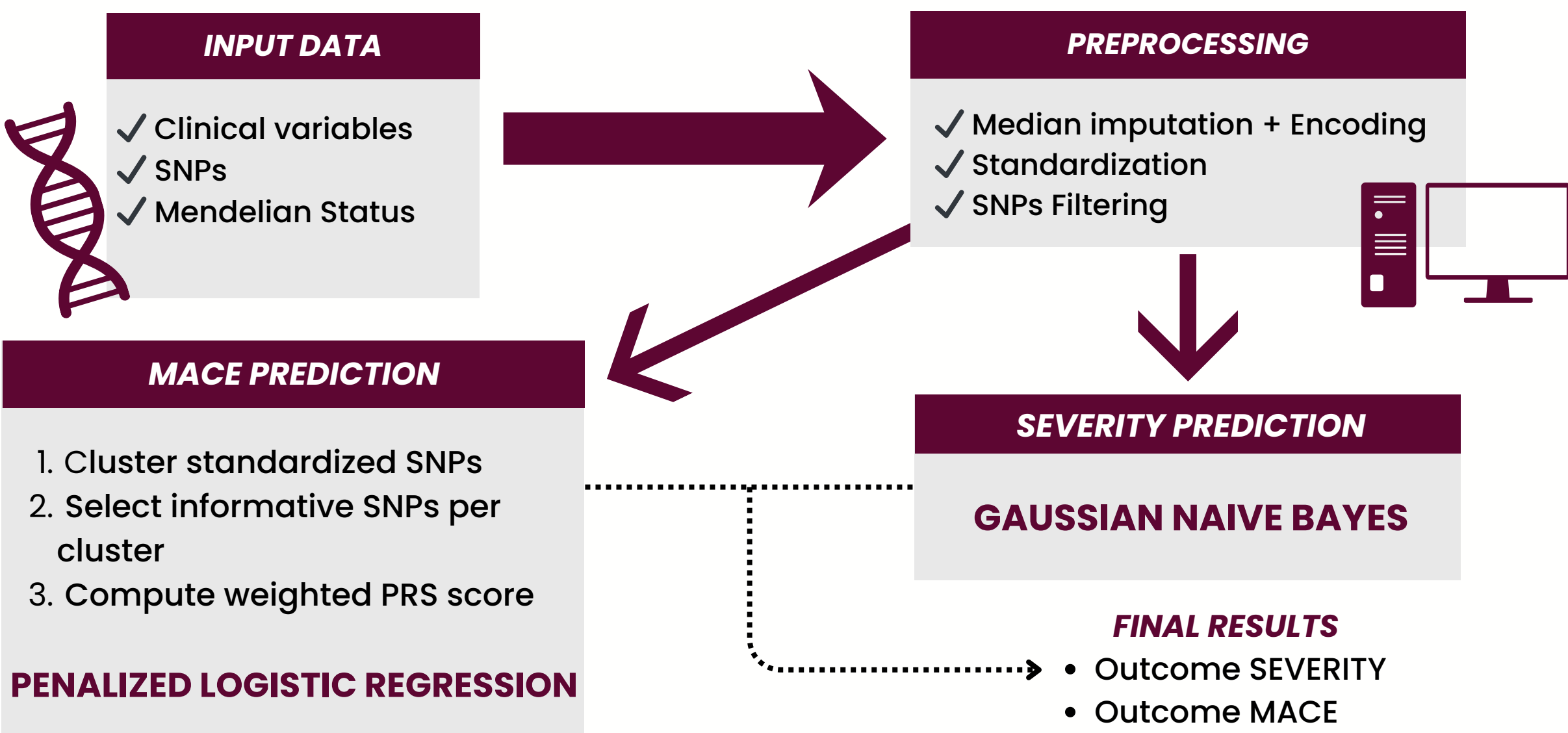


OVERALL ARCHITECTURE OF THE APPROACH

Clinical Preprocessing: Imputes missing values with medians, adds missingness flags, and standardizes features.

Genetic Selection & PRS: SNPs are grouped by similarity using hierarchical clustering to reduce redundancy. The most predictive SNP from each group is chosen via weighted regression, ensuring key priority variants are kept. These are combined into a Polygenic Risk Score (PRS) that quantifies genetic risk for MACE.

MACE Prediction: Penalized logistic regression was used to predict major adverse cardiac events by combining clinical data, genetic status, and the PRS.



The dataset is primarily composed of non-event patients (61.7%), with the remaining 38.3% split between two MACE categories. This imbalance justifies the use of a weighted model to ensure genetic signals from the high-risk minority are not overshadowed by the healthy majority.

INTERPRETATION

The results suggest that combining clinical and genetic information improves prediction performance. Hierarchical clustering reduces redundancy among SNPs, and the PRS summarizes genetic risk in a simple and effective way. Using priority SNPs together leads to better performance, which explains why the final selected SNPs mainly come from this group, as they work better in combination. Overall, the strong performance of the full pipeline shows that genetic information provides complementary value to clinical variables for predicting MACE and may improve risk stratification in hypertrophic cardiomyopathy.